

Diploma Thesis

On the Design Space Exploration of Vibration Analysis Systems

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Predictive Maintenance and Condition Monitoring
Problem Requirements
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Predictive Maintenance and Condition Monitoring

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- Tracks the deterioration of temperature and vibration.
- Alert maintenance team to take action before it is too late.

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Problem Requirements

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- Actual rotating machine for deployment.

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- Development of a secure, authenticated API client for structured data transmission from Raspberry Pi 4 to Supabase.

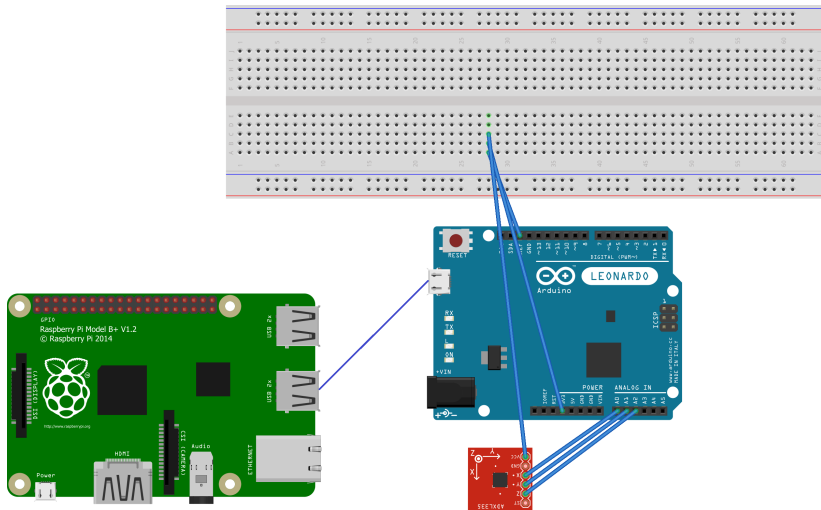
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- Development of a React frontend that requests and visualizes data from the Vercel API.

Predictive Maintenance Tools



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Fourier Transform
Fault Detection

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FFT and DFT

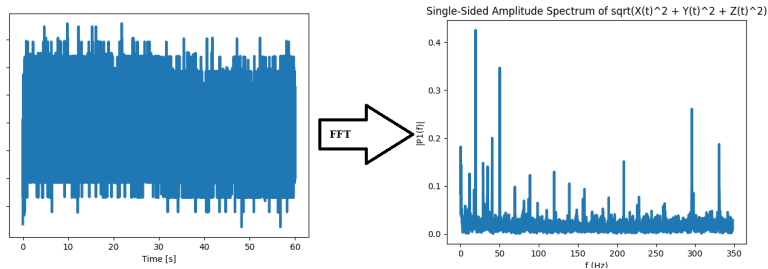


Figure 2: Conversion of the time-domain signal into its frequency-domain power representation using the FFT.

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Seeking a Repeating Pattern and Their Frequency Magnitudes

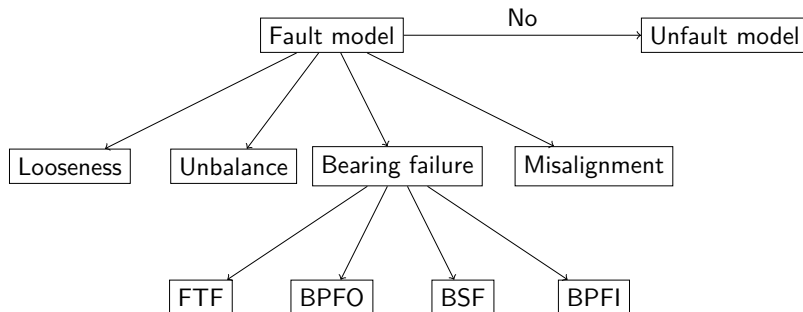


Figure 3: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

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System Architecture and Dataflow

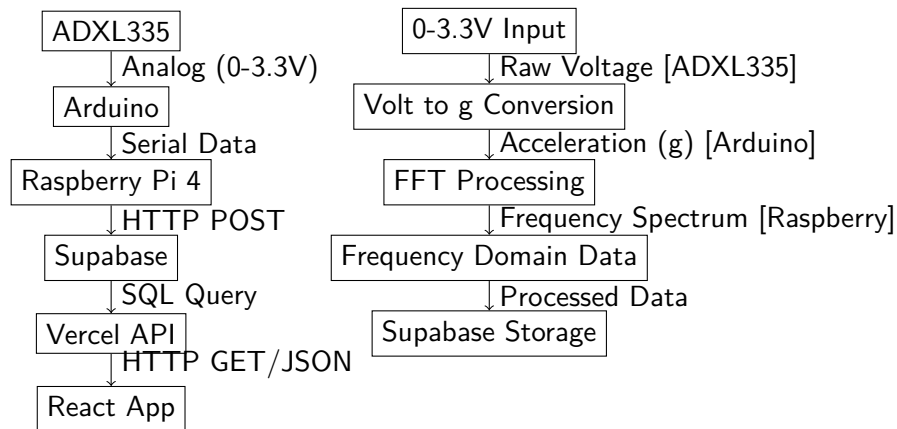


Figure 4: Complete system architecture and data flow and transformation pipeline from physical sensor to web visualization dashboard.

System boundaries

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- Flash memory and SRAM limitations require data to be transmitted in batches.
- The frequency-domain analysis is limited to 350 Hz.
- The measurement time window determines the total number of samples collected.

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Arduino UNO Algorithm

Raspberry Pi Algorithm

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Arduino UNO Algorithm

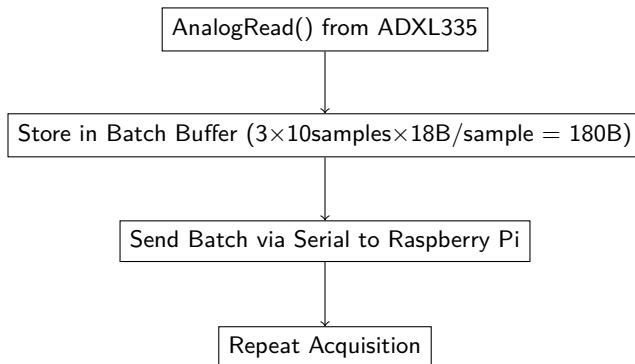


Figure 5: The Arduino UNO Algorithm.

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Raspberry Pi Algorithm

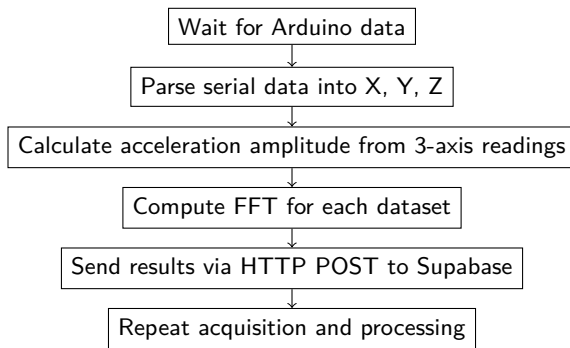


Figure 6: The Raspberry Pi 4 Algorithm: serial data reception, FFT processing, and cloud transmission.

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Vercel and Supabase Algorithmic Flow

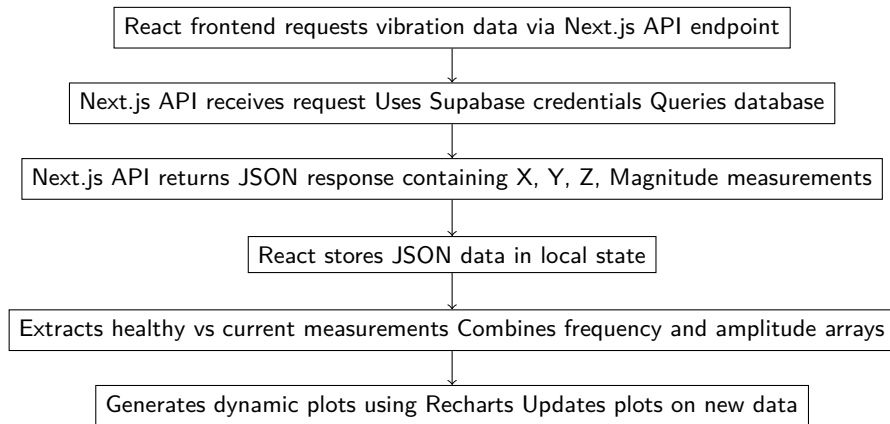


Figure 7: Flow diagram of the Vercel and Supabase algorithm: from frontend request to dynamic visualization.

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Testing on Datasets

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- System performance validated for FFT processing, transmission rates, and visualization updates.

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Testing performance for 1[g] threshold

Table 1: Summary of critical system failures

Fault Condition	Speed	Test Cases	Correct	Incorrect Prediction
Unbalance	50 Hz	11	0	No Alarm
Misalignment	50 Hz	14	0	Correct Alarm
Ball Fault	24 Hz	3	0	No Alarm
Ball Fault	50 Hz	4	0	No Alarm
Inner Race Fault	24 Hz	4	0	No Alarm
Outer Race Fault	24 Hz	3	0	No Alarm

Testing performance for 1[g] threshold

Table 2: Summary of system test results

Task / Condition	Test Cases	Success Rate
Misalignment Prediction	14	100%
Other Bearing Failure	25	0%
Data Transmission (Raspberry → Supabase)	3	100%
Data Transmission (Arduino → Raspberry Pi)	3	100%

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Deployment on working fan for 30secs

Vibration Frequency Dashboard

X Axis

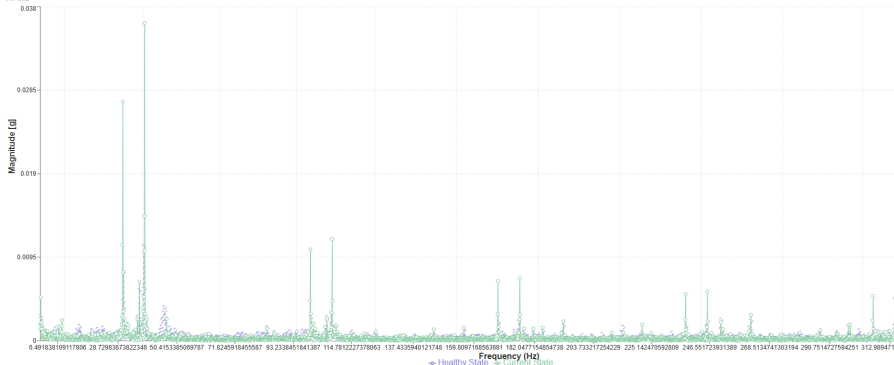


Figure 8: 30 seconds deployment, 500 frequency points and 20684 rows.

Deployment on working fan for 60secs

Vibration Frequency Dashboard

X Axis

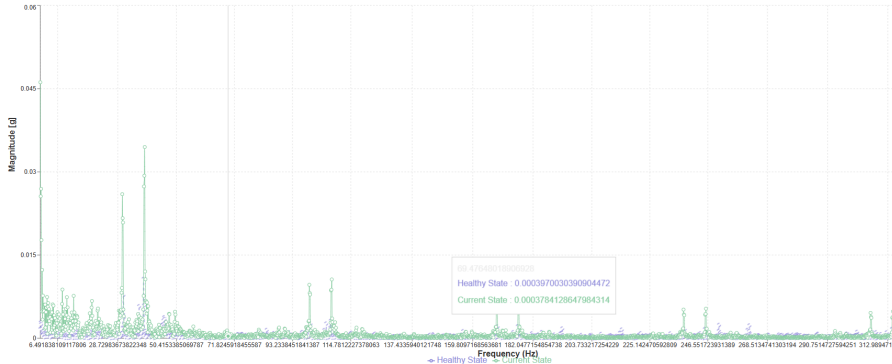


Figure 9: 60 seconds deployment, 2500 frequency points and 41434 rows.

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- Vercel dashboard visualizes frequency spectra interactively for real-time monitoring.